

PEARSON EDEXCEL O LEVEL PHYSICS

Unit 8: Astrophysics and Cosmology

Candidate Name		Date	2026-07-08
Time Allowed	1 Hour	Total Marks	/ 50

Instructions to Candidates

- Answer all questions in the spaces provided.
- Show all stages in any calculations and include appropriate units.
- Calculators may be used.

Questions

Question 1 (Solar System Structures)

(a) The solar system consists of various planetary bodies with differing characteristics. Which of the following correctly pairs an inner terrestrial planet with an outer gas giant?

- A Mars and Mercury
- B Venus and Jupiter
- C Saturn and Neptune
- D Earth and Uranus

Answer: C [1]

(b) Describe the three primary dynamic requirements necessary for a celestial body to be formally classified as a planet.

having their own orbits, having at least one planets that is
orbits their and 2 [3]

(c) State how the orbital speed of a planet varies as its distance from the Sun increases.

Reducing gravitational force cause faster speed to orbit. [1]

Question 2 (Orbital Mechanics)

A moon orbits a distant gas giant planet at an average orbital radius of $r = 4.22 \times 10^8$ meters. The time taken to complete one full orbit is 1.77 Earth days.

(a) Show that the orbital period T of this moon is approximately 1.53×10^5 seconds.

$$1 \text{ day} = 86400 \text{ s}$$

$$1.77 \text{ days} = 1.77 \times 86400 = 1.53 \times 10^5 \text{ s}$$

[2]

(b) Calculate the average orbital speed v of the moon in meters per second (m/s). Use the formula $v = 2\pi r / T$.

$$v = \frac{2\pi \times 4.22 \times 10^8}{1.53 \times 10^5}$$

$$= 1.73 \times 10^4 \text{ m/s}$$

[3]

(c) Explain the fundamental role that gravitational force plays in maintaining this orbital motion, specifying its direction.

Gravitational force provides centripetal force which keeps planets to orbit larger object. As planets orbits, direction of planets is changing.

[3]

Question 3 (Stellar Evolution and Classification)

Stars spend the majority of their operational lives on the Main Sequence before undergoing profound structural shifts governed strictly by their mass.

(a) Contrast the ultimate evolutionary lifecycles of a low-mass star (similar to our Sun) versus a highly massive star after leaving the Main Sequence.

After leaving the Main Sequence, hydrogen is used so stars expands and cool forming red-giant. A low mass star like Sun will cool down and stop glowing and form the planetary nebula and white dwarf. After billions of years, it will stop glowing and form black dwarf. [6]

(b) Explain the structural and physical differences between a *neutron star* and a *black hole*.

Neutron star is smaller size than black hole. Neutron stars emits light while black hole absorb light. [4]

(c) A star's visual properties can be plotted on a Hertzsprung-Russell (H-R) diagram.

(i) State the two primary physical parameters plotted on the axes of an H-R diagram.

luminosity
temperature [2]

(ii) Identify the spectral color corresponding to the hottest main sequence stars compared to the coolest ones.

blue [2]

Question 4 (The Doppler Shift and Redshift)

Light observed from a distant galaxy shows a shift in its spectral lines when compared to a stationary laboratory reference sample.

(a) Define what is meant by the terms *redshift* and *blueshift* in astrophysical observations, clearly linking them to the relative velocity of the source.

Redshift is when the light of the stars moves away from the observer. [4]

Blue shift is when the light of the stars moves toward to the observer.

(b) A specific hydrogen line has a reference wavelength (λ_0) of 656.3 nm. When analyzing light from Galaxy X, the observed wavelength (λ) is measured to be 672.5 nm.

(i) Calculate the change in wavelength ($\Delta\lambda$) for this line.

$$\Delta\lambda = 672.5 - 656.3 = 16.2 \text{ nm} \quad [1]$$

(ii) Using the Doppler redshift equation $\Delta\lambda / \lambda_0 = v / c$, calculate the recessional velocity (v) of Galaxy X. (Speed of light, $c = 3.00 \times 10^8$ m/s).

$$v = \frac{16.2}{656.3} \times 3 \times 10^8 \quad [4]$$
$$= 7.41 \times 10^6 \text{ m/s}$$

Question 5 (Cosmological Theories and Evidence)

The Big Bang Theory asserts that approximately 14 billion years ago, the universe expanded rapidly from an extremely hot, dense point.

(a) State the two paramount pieces of empirical evidence supporting the Big Bang Theory.

higher temperature
higher pressure [2]

(b) Explain comprehensively how the *Cosmic Microwave Background Radiation (CMB)* serves as rigorous verification of the Big Bang model.

Cosmic Background Radiation is founded as electromagnetic^[5] wave when it is detected in anywhere. So, it shows the universe once hot, dense and has expanded and cooled.

